

How to Validate XAI in Longitudinal Studies?

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ABSTRACT

We propose a study design for validating explainable AI (XAI) tools in longitudinal contexts, focusing on ubiquitous and wearable computing. We aim to address unique challenges such as evolving user behavior and temporal dependencies inherent in longitudinal data. The study design includes both general and longitudinal explanation validation, assessing five XAI constructs: trust, satisfaction, actionability, understanding, and reliance through end-user studies. We demonstrate the proposed study design through a working example utilizing two XAI techniques—SHAP-based feature importance and Bayesian counterfactual explanations—on the LAUREATE dataset, which includes machine learning models for estimating weekly affective states of students based on physiological and behavioral data. To the best of our knowledge, this is the first study design that aims to investigate the longitudinal aspects of XAI, by proposing to evaluate the impact of the longitudinal order in which instances and their corresponding explanations are presented to end-users.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing; • Computing methodologies → Artificial intelligence.

KEYWORDS

Explainable AI, Affective Computing, User Studies

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1 INTRODUCTION

As Artificial Intelligence (AI) continues to integrate into ubiquitous technologies, including critical domains such as affective computing [1] and healthcare [2], understanding its decisions becomes a must. Recent regulatory frameworks further underscore AI's transformative power. In the EU, the General Data Protection Regulation (GDPR) [3] set a global standard for data privacy and transparency, and the follow-up AI Act [4] further established comprehensive rules for AI systems, ensuring safety, transparency, and accountability. In the US, an executive order was introduced, emphasizing guiding principles for trustworthy AI [5]. Explainable Artificial Intelligence (XAI) is crucial in this context.

In Ubiquitous Computing, studies have used existing XAI techniques such as LIME [6] or SHAP [7], or have developed new XAI techniques, for example, to explain digital solutions for people with Parkinson's [8] and people with Multiple Sclerosis (MS) [9].

Despite the increased interest in applying XAI, most studies focus on explaining the data or the Machine Learning (ML) models, which is valuable for data and ML engineers. However, considering the central role of end users in Ubiquitous Computing, it is crucial to develop XAI solutions tailored for end users, such as post-hoc personalized explanations. Furthermore, these solutions should be validated through end-user studies to ensure their effectiveness. Despite its growing importance, standardized methods for validating XAI tools are still lacking. Longitudinal use-cases—the majority of cases in Ubiquitous, Pervasive and Wearable Computing—add another layer of complexity.

This paper outlines our ongoing efforts to identify approaches and questionnaires for validating XAI tools through end-user studies. We propose a study design that is applicable for existing longitudinal datasets, measuring five XAI constructs (trust, satisfaction, actionability, understanding, and reliance) in one session per user. For each user, the longitudinal aspects are simulated through the usage of longitudinal datasets, ML models, and explanations with preserved longitudinal order. This approach should help us evaluate the impact of the longitudinal order in which ML instances and their corresponding explanations are presented.

2 XAI USER-STUDIES AND MEASURES

Several studies have tried to address the lack of standardized methods for validating XAI. However, none of these studies take into account the longitudinal component which can be present both in a dataset and in the corresponding explanations. Typical XAI metrics, further explained below, are: *Trust* [11], [12], [14], [15], *Satisfaction* [11], [13], [14], [15], *Actionability* [10], *Understanding* [15], and *Reliance* [16]. These constructs are typically measured using dedicated questionnaires adapted from existing questionnaires originally developed for measuring similar constructs, but in different domains (e.g., see Hoffman et al. [11]).

Trust and Satisfaction – Hoffman et al. [11] presented one of the first studies on metrics for XAI. Their proposed questionnaires for trust and satisfaction are becoming widely used in XAI user studies [11], [12], [13]. These questionnaires reflect the users' trust and satisfaction in the overall decision-making system.

Actionability – Singh et al. [10] introduced a seven-question tool to measure explanation actionability. They evaluated the tool in two separate studies with 182 participants overall. This questionnaire reflects the actionability of each provided explanation.

Understanding – to measure whether explanations help users to understand the decision-making process, Warren et al. [15] divided participants into three groups: counterfactual, causal, and no-explanation control. Each participant completed (i) a training phase with predictions and feedback (explanations for the first two groups, redescription for the control group); and (ii) a testing phase with predictions and no feedback. The differences in accuracy during testing should reflect which type of explanation leads to better understanding.

Reliance – to measure whether users appropriately rely on outputs from a model. That is, users can judge when a model is wrong, and override its decision. Typically, appropriate reliance is defined as accepting an AI prediction when it is correct and rejecting it when it is wrong. Guo et al. [16] show that this definition is not sufficient. They define the *reliance level* as the probability a decision maker chooses the output of the model, *conditioned on them having different predictions*. *Appropriate reliance* is then defined as the behavior that maximizes this probability.

3 MACHINE LEARNING USE CASE

To demonstrate our proposed study design for validating XAI tools in longitudinal use-cases, in this section we describe a typical ML pipeline that includes a longitudinal dataset, ML models for estimating student mood (e.g., weekly mood over a period of 2-3 months), and a corresponding XAI tool for providing explanations.

3.1 Dataset

As a working example, we will use the longitudinal and publicly available LAUREATE dataset [18]. LAUREATE contains data of 42 students collected during 26 classes (a 13-week study), including physiological data (e.g., heart rate and heart rate variability,

sweating rate, skin temperature, and accelerometer data), student performance across the semester, and daily self-reports. The self-reports include positive and negative affect questionnaires, studying hours, physical activity, nutrition habits, and sleep quality. Additionally, all students have filled out the Big Five personality traits questionnaire [20] at the end of the study.

3.2 Machine-Learning Models

In addition to the raw data, the LAUREATE dataset provides baseline ML models for estimating weekly affective states. In this example, we will use the ML models for estimating Negative Affect (NA) [19]. The NA ground-truth levels are calculated from students' self-reported levels of anger, nervousness, stress and worriedness.

As input, the baseline ML models use two modalities. One modality consists of physiological features from data collected 1-2h prior to the self-reported mood levels. The physiological features are: mean skin temperature and sweating rate, number of peaks in the sweating rate signal and their average amplitude, top three frequencies of the accelerometer's magnitude, and ten heart rate variability (HRV) features from the blood volume pulse (BVP) data. The second modality comprises self-reported behavior features: study hours, daily activity, daily breakfast, daily caffeine intake, and daily sleep quality. These features represent the students' behavior in the 24 hours prior to the self-reported mood levels.

Using the two input modalities, the ML models can estimate students' NA levels. For example, in the baseline experiments, personalized random forest (RF) regression models for estimating NA have achieved a Mean Absolute Error of 0.55, which is an improvement compared to a Dummy Regressor that has achieved a Mean Absolute Error of 0.67. More details for the ML models are provided in the LAUREATE study [18].

3.3 XAI Tools

We explore two widely used explainability techniques, SHAP-based feature importance [7], [21], and counterfactual explanations [22], [23]. **SHAP** is a technique that estimates the contribution of each feature to the model's predictions. In the context of the ML models in this study, SHAP values can be used to understand the impact of various physiological and behavioral features on the predicted mood levels. This could provide insights into the model, allowing end users (e.g., the students) to identify key factors that influence their stress and mood.

Counterfactuals offer post-hoc explanations that allow users to discover outcomes for hypothetical scenarios. An example counterfactual explanation for our ML models that estimate affective states (mood) could be: "Your mood could improve if your sleep duration increases by 30 minutes". Such explanations are considered standard for explaining ML models [23]. An example method that could be used to generate high-quality counterfactual explanations is BayCon [22]. BayCon has several characteristics that make it suitable candidate, i.e., it is an efficient Bayesian approach that optimizes multiple objectives for finding good quality counterfactuals, advises only a few and small changes w.r.t the instance to be explained, allows the specification

of immutable features (e.g., a student can't change their age in our use-case), is compatible with continuous and categorical attributes, and works both for regression and classification ML models.

4 PROPOSED VALIDATION APPROACH

Once the explanations are generated (e.g., using SHAP and a counterfactual generator), the next phase is to validate them. We propose a 2-step evaluation study consisting of: (1) a general explanation validation, and (2) a longitudinal explanation validation. The first step evaluates the general quality of the explanations, e.g., by comparing counterfactual vs. SHAP explanations. This step can also be used as a pilot study to test the delivery of the explanations, e.g., counterfactual explanations can be delivered through various visualizations or through text. Once the general quality of the explanations is confirmed, the second step validates the longitudinal component of the explanations. We provide more details in the following two subsections.

4.1 General Explanation Validation

Figure 1 illustrates the proposed methodology for a user study focused on evaluating different explanation types. The study addresses two research questions.

RQ1 – *Is there a statistically significant difference among the three explanation types (counterfactual, feature importance, and no-explanation control) when compared in terms of trust, satisfaction, actionability, reliance, and understanding?*

The second research question targets a component we call *participant-ML data suitability*. In longitudinal studies, ML data corresponds to individuals with certain behavior and personality. For example, in the LAUREATE dataset, all students filled out the “Big Five personality traits” questionnaire and several other questionnaires that describe their behavior (e.g., activity, sleep, nutrition, and smoking habits). This behavior and personality may or may not align with those of participants in the XAI user study. Note that we assume here that the LAUREATE students (participants used to collect ML data) are different from the participants in the XAI validation study (which serves to evaluate the developed XAI tools). This assumption is correct for the majority of longitudinal studies, given that ML models and the corresponding XAI tools are typically developed after the data collection studies. Thus, we state our second research question as follows.

RQ2 – *Is there a positive correlation between participant-ML data suitability and XAI metrics (e.g., trust, satisfaction, actionability, reliance, and understanding)?* Our hypothesis here is that if the behavior of a XAI study participant aligns with a LAUREATE's student behavior (and consequently fits the data used to build the ML models), the XAI study participant might relate better to the provided explanations, compared to other XAI study participants whose behavior is completely different from those in the LAUREATE study.

The proposed methodology involves: (1) Measuring the suitability between participants and ML data. This can be done by asking the participants to fill out the same personality and behavior

questionnaires as those already used in the LAUREATE study. (2) Randomly dividing participants into three groups: Counterfactuals Group, Control Group, and Feature Importance Group. (3) Providing each group with ten training samples and measuring *trust* [11], *satisfaction* [11], *actionability* [10], and *reliance* [16] using the questionnaires from the referenced studies. (4) Providing each group with ten test samples and measuring *understanding* using a similar procedure to that by Warren et al. [15].

Note that the training/test samples, and the corresponding explanations, should come from several ML models (e.g. five modes and four instances per model). We need to test several models because the ML models in the LAUREATE study are personalized, i.e., there are different models for different students. Thus, the explanations and the ML models may vary across different students.

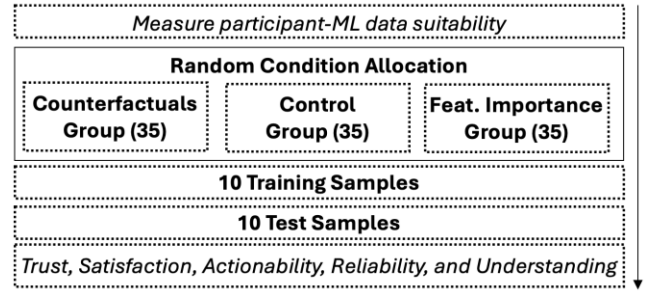


Figure 1. User study for general explanation validation.

4.2 Longitudinal Explanation Validation

This second user study aims to evaluate the impact of the longitudinal order in which ML instances and their corresponding explanations are presented to XAI users in one session. Figure 2 outlines the methodology of the proposed user study that addresses the following research question:

RQ3 – *Does the longitudinal order of the data/explanations positively impact understanding, actionability, trust, satisfaction, and reliance?*

The proposed study design compares two groups. Both groups start with a training phase in which participants are presented with several ML instances from random participants and their corresponding explanations. After the training phase: Group 1, receives explanations for instances which follow a longitudinal order, and Group 2 receives explanations for instances which do not follow a longitudinal order. For example, the LAUREATE dataset contains at least 6-weeks data per student. Thus, we can evaluate XAI quality at several time points. It should be noted that from the mentioned XAI measuring tools (i.e., questionnaires), *actionability* can be measured specifically for each explanation (e.g., Weeks 1, 2, 4, and 5 in the given example); *understanding* is measured on test instances where an explanation is not provided (e.g., Weeks 3 and 6 in the given example); while the rest of the XAI measurements (*trust*, and *satisfaction*) are more suitable to measure the overall system, thus they are applied only at the end of the XAI user study. *Reliance* is measured for each

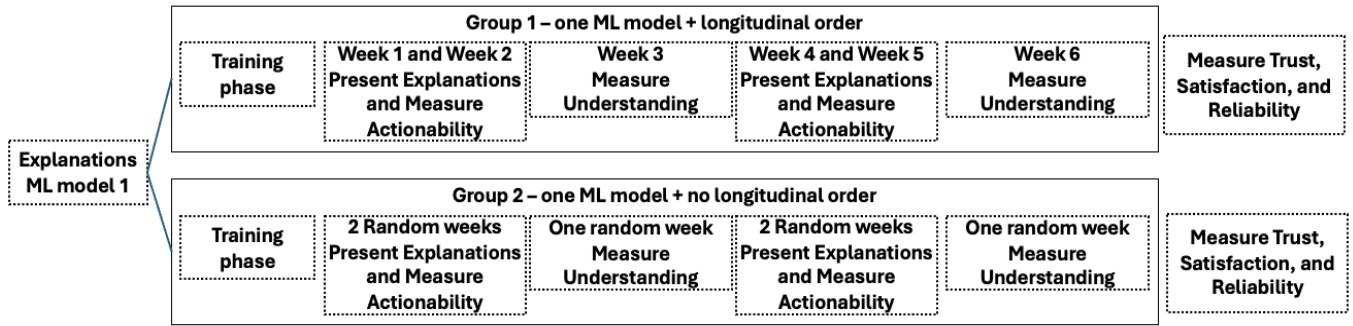


Figure 2. User study for longitudinal explanation validation.

explanation, but the overall appropriate reliance measure is across all of these.

5. DISCUSSION AND CONCLUSION

This paper proposes a new study design for validating XAI tools in longitudinal contexts, particularly within Ubiquitous and Wearable Computing environments. The longitudinal context presents unique challenges due to the evolving nature of user behavior and the temporal dependencies inherent in such data. The proposed study design includes both general and longitudinal XAI validation by measuring constructs such as *trust*, *satisfaction*, *actionability*, *understanding*, and *reliance* through end-user studies. The longitudinal validation further examines how the order and context of explanations over time influence these constructs. In a follow-up study, we will implement the proposed study design to test various XAI tools for explaining ML models, with a focus on affective computing.

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